Fiscal Year:	FY 2020	**Task Last Updated:**	FY 11/18/2020
PI Name:	Mehta, Satish Ph.D.		
Project Title:	Determination of Roles of Microgravity and Ionizing Radiation on the Reactivation of Epstein-Barr Virus		
Division Name:	Space Biology		
Program/Discipline:	SPACE BIOLOGY		
Program/Discipline--Element/Subdiscipline:			
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	(1) Microbiology		
Space Biology Cross-Element Discipline:	(1) Immunology		
Space Biology Special Category:	(1) Cell Culture (2) Translational (Countermeasure) Potential		
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City:	Houston	**State:**	TX
Zip Code:	77058-2720	**Congressional District:**	36
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2012 Space Biology NNH12ZTT001N
Start Date:	10/01/2013	**End Date:**	10/01/2019
No. of Post Docs:	0	**No. of PhD Degrees:**	1
No. of PhD Candidates:	1	**No. of Master' Degrees:**	0
No. of Master's Candidates:		**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA ARC
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Flight Program:			
Flight Assignment:	NOTE: Extended to 9/30/2019 (using 10/01/2019 as end date for reporting purposes) per F. Hernandez/ARC (Ed., 7/10/2018) NOTE: Extended to 9/30/2018 (using 10/01/2018 as end date) per F. Hernandez/ARC (Ed., 9/20/17) NOTE: Extended to 9/30/2017 per F. Hernandez/ARC (Ed., 9/28/16)		
Key Personnel Changes/Previous PI:	no changes		
COI Name (Institution):	Bloom, David Ph.D. (University of Florida) Pierson, Duane Ph.D. (NASA Johnson Space Center) Stowe, Raymond Ph.D. (Microgen, LLC) Wu, Honglu Ph.D. (NASA Johnson Space Center)		
Grant/Contract No.:	Internal Project		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	We will study the effects of simulated microgravity and ionizing radiation on the complex process of virus reactivation that is intimately associated with the way cells respond to external stress. Tools to evaluate the effects of radiation and simulated microgravity will include: both viral and cellular gene expression, viral infectivity and viral titer, and DNA methylation. In addition, this proposal aims to increase our understanding of mechanisms of Epstein-Barr Virus (EBV) reactivation and therefore, has the potential to discover new therapeutic targets for diseases caused in the general population by EBV. Therefore this proposal appropriately addresses: 1. Ground-based Research Designed to Lead to Space Flight; 2. Cell Biology and Microbiology: This section of the NASA Research Announcement (NRA) solicits NOVEL ground-based experiments that use cell cultures, tissue cultures, microbiological organisms, or small invertebrate organisms to elucidate the basic mechanisms living systems use to respond to changes in gravity, i.e., how are changes sensed and transduced to alter gene expression and the structure and function of the musculoskeletal system. The reactivation of all herpes viruses is known to involve 2 components or stages: the initial initiating molecular events that cause reactivation of the latent genomes to progress to the production of reactivating virus, and then the propagation of these reactivating viruses to produce clinical disease. The second part of this process undoubtedly involves overcoming components of the acquired immune response if clinical disease is to result. In order to understand reactivation from a mechanistic point of view, these two processes need to be examined independently. In the proposed studies, we will be focusing only on the initial events that microgravity and ionizing radiation initiate in latent cells to induce the first stage of (molecular) reactivation. EBV reactivation may not be completely homologous to Varicella zoster virus (VZV) and Cytomegalovirus (CMV) reactivation. The important point here though is that by focusing on EBV, which has a well-established cell culture system of latency, we can study the mechanistic effects of microgravity and ionizing radiation in inducing EBV reactivation. This alone will provide critical and important information regarding the effects of these inducers on EBV reactivation in humans. Clearly, after these molecular events are established in EBV, possible parallel mechanisms in VZV and CMV can be investigated in future studies.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	This proposal aims to increase our understanding of mechanisms of Epstein-Barr Virus (EBV) reactivation and therefore has the potential to discover new therapeutic targets for diseases caused in the general population by EBV.
Task Progress:	Final Report, June 16, 2020 SUMMARY We studied the effects of simulated microgravity and ionizing radiation on the complex process of virus reactivation that is intimately associated with the way cells respond to external stress. Tools to evaluate the effects of radiation and simulated microgravity included: both viral and cellular gene expression, viral infectivity, and viral titer. In addition, this study aimed to increase our understanding of mechanisms of EBV reactivation and therefore had the potential to discover new therapeutic targets for diseases caused in the general population by EBV. Therefore this proposal appropriately addressed ‘Pathogen infectivity and host immunity’ using human blood cells (B lymphocyte cell line infected with EBV) as one of the targeted research topics by the Heart, Lung and Blood Institute (NHLBI) described on p-6 of PAR-09-120: Biomedical research on the International Space Station (Biomed –ISS) (UH2). The reactivation of all herpesviruses is known to involve 2 components or stages: the initial initiating molecular events that cause reactivation of the latent genomes to progress to the production of reactivating virus, and then the propagation of these reactivating viruses to produce clinical disease. The second part of this process undoubtedly involved overcoming components of the acquired immune response if clinical disease is to result. In order to understand reactivation from a mechanistic point of view, these two processes were examined independently. In the proposed studies, we focused only on the initial events that microgravity and ionizing radiation initiate in latent cells to induce the first stage of (molecular) reactivation. EBV reactivation may not be completely homologous to VZV and CMV reactivation. The important point here though is that by focusing on EBV, which has a well-established cell culture system of latency, we can study the mechanistic effects of microgravity and ionizing radiation in inducing EBV reactivation. This alone will provide critical and important information regarding the effects of these inducers on EBV reactivation in humans. Clearly, after these molecular events are established in EBV, possible parallel mechanisms in VZV and CMV can be investigated in future studies. Among the many stressors astronauts are exposed to during spaceflight, cosmic radiation may lead to various serious health effects. Specifically, space radiation may contribute to decreased immunity, which has been documented in astronauts during short and long-duration missions, as evidenced by several changes in cellular immunity and plasma cytokine levels. Reactivation of latent herpes viruses, either directly from radiation of latently infected cells and/or from perturbation of the immune system, may result in disease in astronauts. Epstein-Barr virus (EBV) is 1 of the 8 human herpes viruses known to infect more than 90% of human adults and persists for the life of the host without normally causing adverse effects. Reactivation of several latent viruses in astronauts is well-documented, although the mechanism of reactivation is not well understood. We studied the effect of 4 different types of radiation, (1) 137Cs gamma rays, (2) 150-MeV protons, (3) 600 MeV/n carbon ions, and (4) 600 MeV/n iron ions on the activation of lytic gene transcription and of reactivation of EBV in a latently infected cell line (Akata) at doses of 0.1, 0.5, 1.0, and 2.0 Gy. The data showed that for all doses used in this study, lytic gene transcription was induced and median viral loads were significantly higher for all types of radiation than in corresponding control samples, with the increases detected as early as 4 days post-exposure and generally tapering off at later time points. The viability and size of EBV-infected Akata cells were highly variable and exhibited approximately the same trend in time for all radiation types at 0.1, 0.5, 1.0, and 2.0 Gy. This work shows that reactivation of viruses can occur due to the effect of different types of radiation on latently infected cells in the absence of changes or cytokines produced in the immune system. In general, gamma rays are more effective than protons, carbon ions, and iron ions in inducing latent virus reactivation, though these high energy particles did induce more sustained and later reactivation of EBV lytic gene transcription. These findings also challenge the common relative biologic effectiveness concept that is often used in radiobiology for other end points.
Bibliography Type:	Description: (Last Updated: 04/15/2025)

Books/Book Chapters	Stowe RP, Pierson DL, Mehta SK. "Stress, spaceflight, and latent herpes virus reactivation." in "Stress Challenges and Immunity in Space: From Mechanisms to Monitoring and Preventive Strategies." Ed. A. Choukèr. Cham: Springer International Publishing, 2020. p. 357-372. https://doi.org/10.1007/978-3-030-16996-1_19 , Jan-2020